

Who Hedges the Barrel?

Sovereign Oil Hedging and Petro-Currency Reactions to OPEC News

Dissertation outline | Will McGowan, BA (Hons) Economics, Strathclyde | For Dr Luigi Gifuni | October 2026

Question. When an OPEC announcement moves the oil price, how much do oil exporters' currencies move, and does a government that hedges the country's oil exposure change the shape of that reaction?

Three predictions, written down before testing

1. Norway (pre-announced krone conversions for the oil fund): a smaller oil beta than unhedged Canada.
2. Russia (2017–22 budget rule, which bought foreign currency when oil was above a cut-off price and sold it below): a reaction that changes at the cut-off.
3. Mexico (annual put-option hedge): a one-sided reaction, weaker to bad oil news than to good.

Why it is new. Känzig (2021) identifies oil supply news from OPEC announcements, and studies of exporters' currencies explain their reactions with export exposure or monthly shocks. To my knowledge, none links currency reactions on OPEC days to the hedging institutions themselves. I will confirm this against the closest recent paper, Rüth et al. (2026), in the first week.

Data (all free). Känzig's OPEC-announcement oil supply surprises (daily, since 1983) and Degasperi's cleaned series; daily exchange rates from FRED and the ECB; Norges Bank's monthly krone-conversion announcements; the Russian Finance Ministry's monthly foreign-currency purchase announcements (2017–22); Mexico's hedge strikes and costs (Finance Ministry reports; Ma and Valencia, IMF Working Paper 18/35).

Method (MATLAB)

- **Baseline:** each currency's same-day beta to the OPEC surprise, against the dollar and the euro, with a dollar factor removed; HAC and block-bootstrap inference.
- **Norway vs Canada:** the beta interacted with that month's announced conversion flow, plus an event study of the conversion announcements themselves.
- **Russia:** difference-in-differences across the rule's dated regimes, with CAD and NOK as controls; data after February 2022 excluded.
- **Mexico:** separate betas for positive and negative surprises, compared across hedged years (2001–17, while hedge volumes were disclosed).
- **Tradability:** one pre-registered OPEC-day trading rule, judged after costs with a White reality check and Kelly sizing.
- **Robustness:** Degasperi's series, two-day windows, and VIX and S&P 500 controls.

Fit with your brief. Topic 3, how institutions respond to projections of commodity prices: each rule is set from an oil-price assumption or a revenue projection. The design follows the event-study approach of your Brexit paper, with the out-of-sample discipline of your forecasting work.

Risks and fallback. Few OPEC days fall inside each regime, so the Russia and Mexico tests may lack power; they are pre-registered and reported either way. If they are inconclusive, the Norway–Canada comparison and a currency-by-currency table of OPEC-day betas still stand on their own.

Timeline. October–November: data and baseline betas. December–January: Norway–Canada and Russia. February: Mexico and the trading test. March: robustness and writing up.

Where your advice would matter most. Combining Känzig's and Degasperi's series, the choice of event window, and inference with so few OPEC days in each regime.